Quantitative Portfolio Optimization

- Markowitz Mean-Variance Framework
- Capital Asset Pricing Model (CAPM)
- Monte Carlo Simulations
- Efficient Frontier Analysis

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Comparison between CAPM and Domestic Tangency Portfolio Weights

The results from the Markowitz optimization, which defines risk as **total volatility** and **correlation**, drove the model to build a highly concentrated domestic tangency portfolio to achieve the maximum Sharpe Ratio (1.44). This aggressive portfolio was heavily weighted in **COFORGE (50.16%)** and **ASTRAZEN (47.53%)**, implying this pair had the optimal blend of high historical returns and/or beneficial (low) correlation to lower portfolio volatility. Consequently, **GODREJCP (0.62%)**, **BRIGADE (1.02%)**, and **GUJGASLTD (0.67%)** were deemed inefficient and almost entirely excluded.

In stark contrast, the Capital Asset Pricing Model (CAPM) defines risk purely as **systematic risk (Beta)**. Our analysis identified a critical **negative Market Risk Premium (MRP)** for this period (Nifty 50 return 5.69% < risk-free rate 6.66%), which inverted the typical CAPM relationship. This caused the model to penalize the highest-risk stock, **BRIGADE (Beta 1.32)**, with the *lowest* required return (5.38%), while rewarding the most defensive stock, **GODREJCP (Beta 0.41)**, with the *highest* required return (6.26%). This reveals the core conflict: the backward-looking Markowitz model prioritized the highest-performing assets (COFORGE/ASTRAZEN), while the CAPM, reacting to the negative market, identified the defensive, low-beta **GODREJCP** as the most valuable asset. In order to quantitatively acknowledge the findings, kindly refer to *table 1* and *table 2* below.

Table 1: Weights of Securities in Portfolios along with their CAPM Beta

Security	GODREJCP	ASTRAZEN	COFORGE	BRIGADE	GUJGASLTD	NVDA
Domestic Tangency Weights	0.62%	47.53%	50.16%	1.02%	0.67%	-
International Tangency Weights	1.60%	33.63%	46.91%	4.12%	0.37%	13.37%
CAPM Beta	0.41	0.60	0.79	1.32	0.94	-

Comparison between Domestic Tangency and Index (Passive Portfolio)

The domestic tangency portfolio, with its Sharpe ratio of **1.44**, is significantly greater than the index (passive portfolio), which had a negative risk-adjusted return for the period. This indicates our actively constructed portfolio is far superior to the passive index, and investors would strongly prefer it. In order to prove the aforementioned findings, kindly refer to *table 3* below.

Table 2: Important Parameters of Securities

Security	GODREJCP	ASTRAZEN	COFORGE	BRIGADE	GUJGASLTD
Annualized Return	61.39%	11.14%	47.58%	0.01%	-23.08%
Annualized Risk	44.77%	42.25%	36.06%	28.21%	31.76%
Sharpe Ratio	1.2225	0.1060	1.1349	-0.2357	-0.9364

Table 3: Important Parameters of Portfolios

Security	Domestic Tangency	International Tangency	Index (NIFTY50)
Expected Return	50.49%	45.95%	5.69%
Annualized Risk	30.5%	26.38%	14.1%
Sharpe Ratio	1.44	1.49	-0.0691

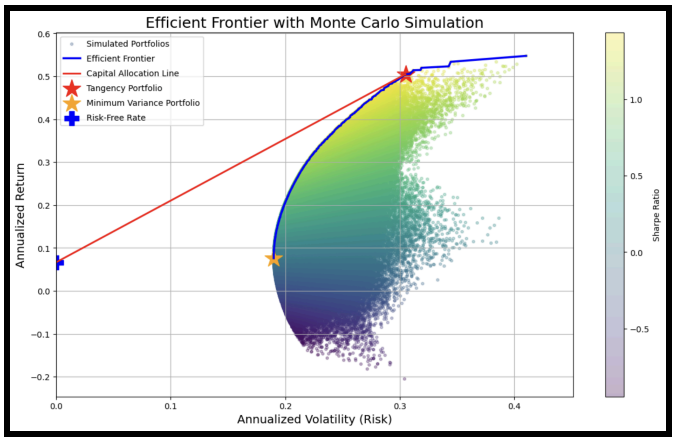
Comparison between Domestic and International Tangency Portfolios

The international tangency portfolio, with a Sharpe ratio of **1.49**, is marginally superior to the domestic tangency portfolio, which had a Sharpe ratio of **1.44**. In addition to this, the international portfolio carries a significantly reduced risk (**26.38%**) compared to the purely domestic portfolio (**30.50%**). This makes it evident that adding an international security (NVDA) provided powerful diversification benefits, as its low correlation to the domestic assets lowered the portfolio's total volatility. Hence, the international tangency portfolio would be the more attractive investment, as it provides a superior risk-adjusted return with less risk. In order to acknowledge the aforementioned conclusions, kindly refer to *Table 3* above.

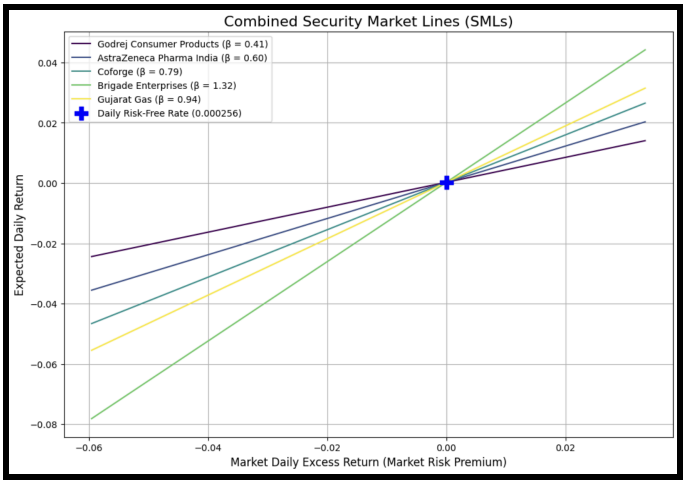
Comparison between International Tangency and Index (Passive Portfolio)

The international tangency portfolio, which yielded a Sharpe ratio of **1.49**, is significantly and overwhelmingly superior to the index (passive portfolio), which had a negative Sharpe ratio of **-0.07** for the period. This massive gap in risk-adjusted returns indicates that our optimized 6-stock portfolio is a far more efficient investment, and investors would strongly prefer it over a passive index-tracking strategy. In order to quantitatively acknowledge the aforementioned findings, kindly refer to *Table 3* above.

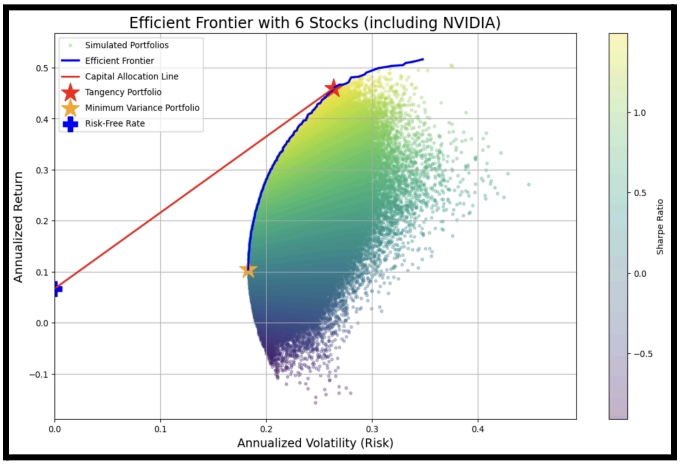
Appendix



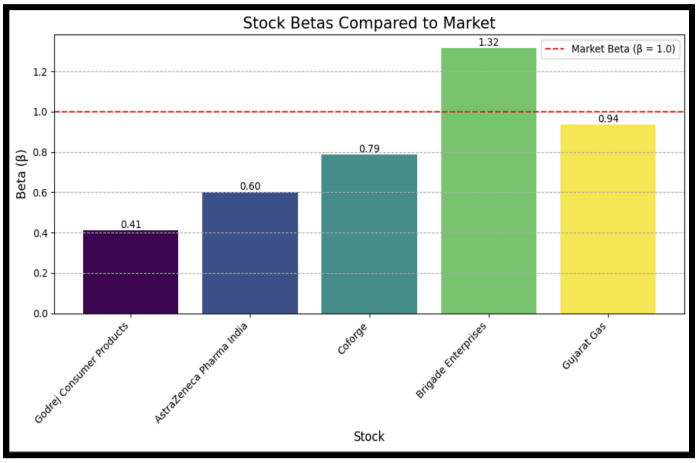
Efficient Frontier for Domestic Portfolio



SML Lines for all Domestic Securities



Efficient Frontier for International Portfolio



Beta of Securities Compared to Market

References

1. NSE India Official Page (https://www.nseindia.com/)
2. Investing.com (https://www.investing.com/)
3. NASDAQ Official Page (https://www.nasdaq.com/)
4. Group-15_SAPM_Assignment_Excel File (for extracting the tables)
5. Group-15_SAPM_Assignment_Code File (for calculations and visual representations of the results)